CSCI 4160 – Interactive Media

Final Project - Void Surge - GDD

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Void Surge - Game Design Document

Game Overview

- **Title:** Void Surge
- **Genre:** Top-down arena survival / roguelite
- **Engine:** Godot 4.6.2 (GDScript)
- **Platform:** Windows (PC), exportable to Mac/Linux
- **Target Audience:** Players who enjoy fast-paced action roguelites (Vampire Survivors, Nuclear Throne, Enter the Gungeon). Suitable for casual and mid-core gamers aged 13+.
- **Session Length:** 10–15 minutes per run

Core Gameplay Loop

1. **Spawn** into a contained arena
2. **Survive an enemy wave**: shoot, dodge, and dash to stay alive
3. **Manage heat**: shooting builds heat; high heat boosts damage but overheating burns you and locks your weapon
4. **Clear the wave**: all enemies must be destroyed
5. **Choose an upgrade**: pick 1 of 3 random upgrades to strengthen your build
6. **Repeat**: waves escalate in enemy count, type variety, and stat scaling
7. **Win or die**: survive all 10 waves to win, or die trying

Mechanics (MDA - Mechanics)

Controls

Action	Keyboard/Mouse	Gamepad
Move	WASD / Arrow Keys	Left Stick
Aim	Mouse cursor	Right Stick (aim toward)
Shoot	Left Mouse Button	RT
Dash	Space / Right Mouse Button	A / South Button
Pause	Escape	Start

Core Systems

1. Heat System:

- Every shot adds heat (default 8 per shot, max 100)
- Heat decays passively over time (default 25/sec).
- **Damage bonus:** Bullets deal up to +50% damage at max heat (scales linearly with heat %).
- **Overheat penalty:** Reaching 100 heat deals 15 self-damage and locks weapons for 2.5 seconds.
- **Strategic tension:** Players must decide whether to "run hot" for damage or play it cool for safety.

2. Dash System:

- Press dash to lunge in the movement direction (or toward cursor if stationary).
- Grants invincibility frames during the 0.15s dash.
- Dash has a cooldown (default 1.2s).
- **Dash damage:** Passing through enemies during a dash deals damage (default 30).
- Creates a viable melee/dash-focused playstyle as an alternative to pure ranged.

3. Weapon/Shooting

- Hold shoot for automatic fire at the fire rate interval.
- Bullets travel in the aimed direction with configurable speed.
- **Spread shot:** Upgrades add additional projectiles in a fan pattern.
- **Piercing:** Upgraded bullets pass through multiple enemies.

4. Upgrade System

- After clearing each wave, the game pauses and presents 3 random upgrades.
- 11 total upgrades in the pool (can be picked multiple times, stacking).
- Categories: damage, fire rate, heat management, dash power, survivability, movement.
- Player choices create emergent build identities (see Dynamics).

Core Resources

- **Health:** Starts at 100. Lost to enemy contact, enemy bullets, and overheating. Regenerated only via upgrades.
- **Heat:** 0–100 gauge. Rising heat = more damage but approaching the overheat threshold.
- **Dash charge:** Binary cooldown; available or recharging.

Progression

- **In-run:** Upgrade choices between waves (9 upgrades across a full run).

- **Wave scaling:** Enemy health, damage, and variety increase each wave.

Dynamics (MDA — Dynamics)

Expected Strategies & Emergent Builds

Build Archetype	Key Upgrades	Playstyle
Glass Cannon	Power Shot, Rapid Fire, Pyromaniac	Maximize DPS; accept overheat risk; rely on positioning
Cold Efficiency	Coolant System, Piercing Rounds, Scatter Cannon	Sustained fire without overheating; crowd control
Dash Assassin	Blade Dash, Quick Step, Thrusters	Close-range dash damage; weapon is secondary
Tank	Reinforced Hull, Auto Repair, Move Speed	Outlast enemies; absorb hits; slow and steady

Most runs will be hybrids, the randomized upgrade pool forces adaptation.

Mechanic Interactions

- **Heat × Combat:** High heat = high reward (damage bonus) but high risk (overheat lockout + self-damage). This creates a constant micro-decision during combat.
- **Heat × Dash:** When overheated, the player cannot shoot, dashing becomes the only offensive and defensive option during lockout. Dash upgrades directly mitigate overheat punishment.
- **Upgrades × Enemy Types:** Chasers pressure the player to move and dash; Shooters demand positioning; Dashers require timing. Upgrade choices shift which enemy types feel threatening.
- **Spread × Pierce:** Combining these creates wide area damage, fundamentally changing how the player engages groups.

Difficulty Curve

- **Waves 1-3:** Chasers only, low counts. Teaches movement, shooting, heat awareness.
- **Waves 4-6:** Shooters added. Forces positioning and dodging bullets while managing heat.
- **Waves 7-9:** Dashers added. Fast enemies demand dash usage and spatial awareness.
- **Wave 10:** All types, high counts. Tests the full build the player has assembled.

Aesthetics (MDA - Aesthetics)

Intended Feelings

- **Challenge:** Escalating waves that test mastery of movement and heat management.
- **Discovery:** Each run's random upgrades create a different build and experience.
- **Sensation:** Fast, responsive controls with immediate visual feedback (hit flashes, screen shake, particle bursts, color-shifting heat indicator on the player).

What Creates Them

- **Challenge** ← wave scaling + overheat punishment + enemy variety forcing adaptation
- **Discovery** ← randomized upgrade selection + stacking synergies
- **Sensation** ← screen shake on damage + white flash on enemy hits + player color shifting cyan→red with heat + death particle bursts + OVERHEATED warning text

Visual Style

- Dark void background (near-black with subtle blue tint)
- Geometric shapes: triangle (player), triangle (chaser), square (shooter), diamond (dasher)
- Color-coded: cyan (player), red (chaser), orange (shooter), purple (dasher)
- Player body color shifts from cyan to orange-red as heat increases
- Neon-accented arena border with corner highlights

Content Plan (Final Build)

Content	Count	Details
Enemy types	3	Chaser, Shooter, Dasher
Waves	10	Escalating composition and stats
Upgrades	11	Covering damage, defense, mobility, heat, and utility
Screens	5	Main menu, HUD, pause, game over, victory
Arena	1	Fixed-size bordered arena

Implementation Plan

Key Systems

1. **GameManager (Autoload):** Global state singleton - score, wave, player stats, upgrade application.
2. **Wave Manager:** Spawns enemies per wave definition, tracks remaining enemies, signals wave clear.
3. **Player Controller (CharacterBody2D):** Movement, shooting, dash with i-frames, heat management, damage/flash feedback.
4. **Enemy Base Class:** Shared health/damage/death logic. Subclasses implement AI: Chaser (seek), Shooter (kite + fire), Dasher (circle + charge).
5. **Upgrade Panel (CanvasLayer):** Pauses game, presents 3 random cards, applies selection to GameManager stats.
6. **HUD:** Real-time bars for health, heat, dash cooldown; wave/score labels; wave banner animation.

Technical Risks & Mitigations

Risk	Mitigation
Bullet performance with spread + pierce	Object pooling if needed; bullets self-destruct on lifetime/distance
Enemy pathfinding edge cases	Simple steering behaviors, no navmesh; arena clamping prevents stuck enemies
Overheat feels unfair	Self-damage is moderate (15); clear visual/audio warning at high heat; lockout duration tuned to 2.5s
Upgrade balance	Upgrades stack linearly with caps; playtested to ensure no single upgrade trivializes the game